

Unsupervised Machine Learning for Network Traffic Clustering: A Comparative Analysis of Lab vs. Real-World Telemetry

Stancu Vlad-Andrei

Universitatea de Vest din Timisoara / Intelligent Software Robotics

June 20, 2026

Abstract

This project evaluates the efficacy of unsupervised machine learning, specifically the DBSCAN clustering algorithm when applied to an anomalous network traffic without prior labeling. By comparing a historical, controlled baseline dataset (CIC-IDS-2017) against modern, real-world honeypot telemetry (LUFlow 2022), this study demonstrates the severe impact of feature drift and environmental noise on the algorithm's performance. The results indicate that while density-based clustering effectively isolates deterministic attack scripts in a controlled environment (achieving a Silhouette Score of 0.29 across 16 highly specific micro-clusters), it struggles with the chaotic nature of modern threat environments. In real-world telemetry, overlapping features cause the algorithm to group anomalies into fewer, massive macro-clusters, highlighting the limitations of traditional spatial clustering in modern Intrusion Detection Systems (IDS).

Contents

1	Introduction	3
2	Description of the Datasets	3
2.1	CIC-IDS-2017 (Baseline)	3
2.2	LUFlow 2022 (Real-World)	3
3	Description of the Processing Flow	4
3.1	Data Sanitization and Scaling	4
3.2	Clustering Methodology	4
4	Presentation of Results	4
4.1	Visual Dimensionality Reduction (PCA)	4
4.2	Quantitative Evaluation and The Silhouette Paradox	5
5	Conclusions	5

1 Introduction

With the rapid evolution of cyber threats, traditional signature-based Intrusion Detection Systems (IDS) frequently fail to detect zero-day vulnerabilities and polymorphic malware. To address this, network security researchers have been increasingly leaning towards machine learning to detect anomalies based on behavioral deviations rather than static signatures.

To explore the viability of this approach, this report analyzes two distinct network datasets: the highly cited CIC-IDS-2017, representing a sterile laboratory environment with deterministic attack scripts, and the LUFlow 2022 dataset, representing active, real-world honeypot telemetry [2]. Previous studies utilizing CIC-IDS-2017 have largely relied on supervised classification methods, often achieving artificially high accuracy due to the tightly controlled nature of the traffic [1].

This report proposes an unsupervised approach using Density-Based Spatial Clustering of Applications with Noise (DBSCAN) [3]. By relying purely on spatial density rather than pre-defined labels, we aim to mathematically isolate anomalous traffic. The rest of this report describes the two datasets, explains how the data was cleaned, and compares the clustering results between the 2017 lab baseline and the 2022 real-world traffic.

2 Description of the Datasets

This study utilizes two distinct network traffic datasets. To maintain computational parity and allow for a controlled scientific comparison, uniform subsets of 80,000 records were extracted from both data sources.

2.1 CIC-IDS-2017 (Baseline)

The baseline dataset consists of traffic captured over five days of simulated, scripted network attacks. The feature matrix contains 78 continuous numerical attributes generated by the CICFlowMeter extraction tool (e.g., Flow Duration, Packet Lengths, Byte Rates). The dataset showcases a severe class imbalance, typical of lab environments, dominated by Benign traffic alongside specific attack vectors like DoS Hulk, PortScan, and DDoS.

Preliminary statistical analysis (mean, standard deviation, min, max) was conducted on both datasets. For example, the baseline dataset showed extreme variance in flow duration and byte rates, necessitating the aggressive standardization pipeline detailed in Section 3.1.

2.2 LUFlow 2022 (Real-World)

The modern dataset consists of traffic captured via global honeypots [2], yielding 15 continuous numerical features after stripping string-based metadata such as IP addresses. Unlike the baseline, the LUFlow dataset target distribution is highly balanced, containing nearly a 50/50 split of normal and anomalous traffic (40,072 benign vs. 39,928 outliers), reflecting the constant bombardment of automated scanners and bots on the modern internet.

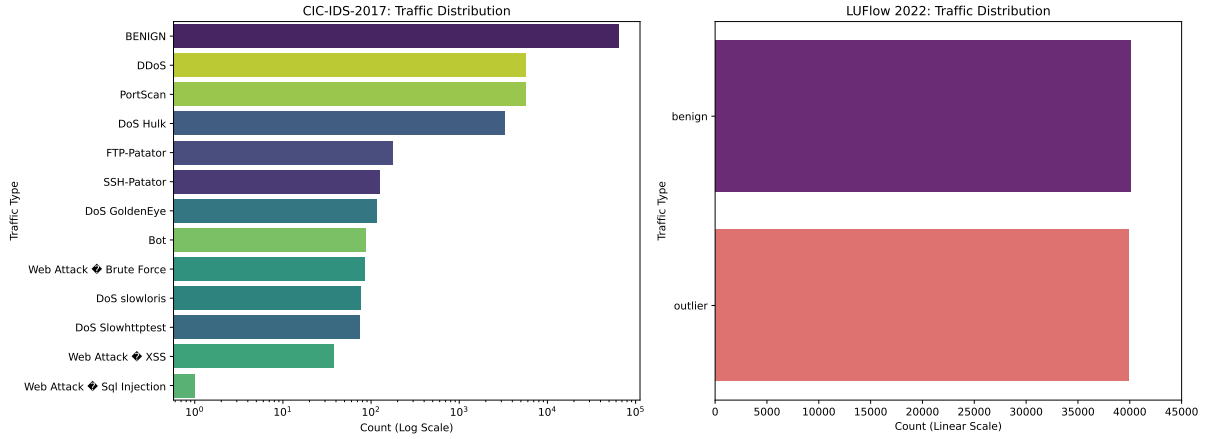


Figure 1: Target class distributions showcasing the severe imbalance in the 2017 baseline (log scale) versus the balanced nature of the 2022 real-world telemetry (linear scale).

3 Description of the Processing Flow

DBSCAN relies entirely on calculating Euclidean distances between data points. Therefore, the data pipeline was strictly engineered to sanitize and standardize the feature space.

3.1 Data Sanitization and Scaling

Raw network telemetry frequently contains calculation artifacts. Infinite values (a known bug in flow extraction tools when packet duration is instantaneous) were converted to NaN and subsequently purged from the matrices. This resulted in a negligible data loss (less than 0.1% for 2017 and approximately 8% for the noisier 2022 dataset).

Following sanitization, standard scaling was applied. To prevent high-variance features (such as 'Total Length of Fwd Packets') from mathematically dominating the distance calculations, every numerical column was standardized to have a mean of zero and a standard deviation of one ($\mu = 0, \sigma = 1$).

3.2 Clustering Methodology

DBSCAN [3] was selected due to its inherent ability to classify sparse, low-density data points as anomalies (assigned the label -1). To ensure a fair comparison, the algorithm was initialized with identical hyperparameters for both datasets: **eps=2.5** (the maximum distance between two samples to be considered neighbors) and **min_samples=15** (the core point threshold).

It is important to note that DBSCAN computes a distance matrix with an $O(n^2)$ memory and time complexity. To prevent memory overflow while maintaining statistical significance, the clustering algorithm was fitted on a subset of 20,000 records for both datasets.

4 Presentation of Results

The performance of the DBSCAN algorithm varied drastically between the controlled and the real-world data, revealing critical insights into the nature of modern cyber threats.

4.1 Visual Dimensionality Reduction (PCA)

To visually interpret the high-dimensional network structures (78 and 15 features, respectively), Principal Component Analysis (PCA) was utilized to perform dimensionality reduction, projecting the complex data onto a 2D plane for visual inspection.

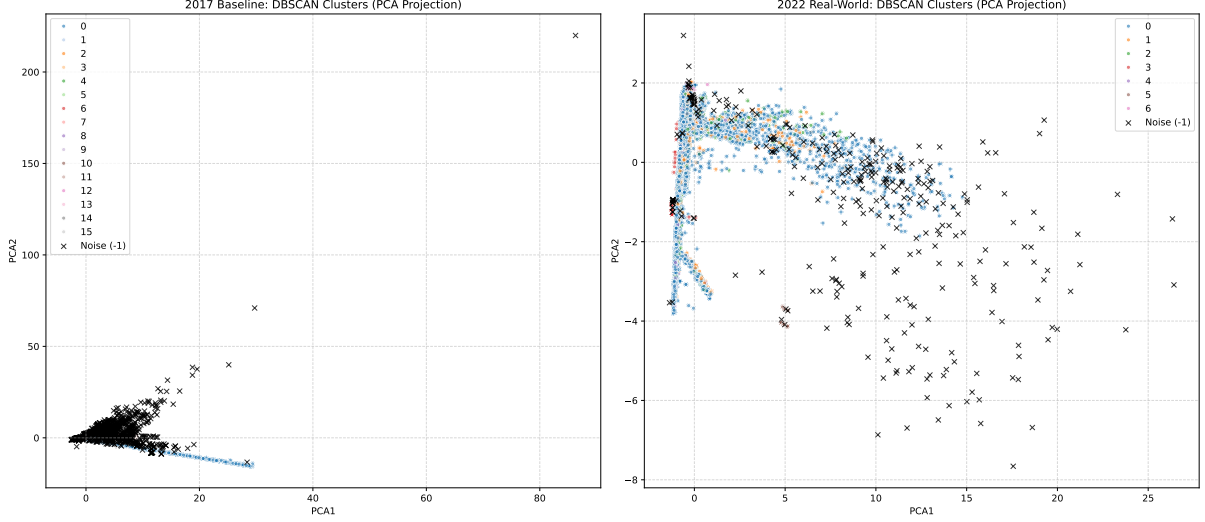


Figure 2: PCA projection of DBSCAN clusters. The 2017 baseline (left) exhibits rigid, deterministic attack patterns, while the 2022 telemetry (right) shows chaotic, overlapping features typical of real-world evasion tactics.

As shown in Figure 2, the 2017 baseline forms rigid, distinct linear groupings, proving that the attacks were generated by highly predictable, automated scripts. In contrast, the 2022 data is smeared in a chaotic arc with widely dispersed anomalies, illustrating the overlapping features of modern evasion techniques.

4.2 Quantitative Evaluation and The Silhouette Paradox

Using the Silhouette Score to mathematically measure cluster cohesion and separation (excluding noise points), the algorithm yielded the following metrics:

- **2017 Baseline:** 16 distinct clusters, 5.47% Noise, Silhouette Score: **0.2921**
- **2022 Real-World:** 7 distinct clusters, 1.61% Noise, Silhouette Score: **0.3735**

While the quantitative Silhouette Score suggests the 2022 dataset has higher structural clustering, a critical visual analysis reveals this to be a mathematical paradox. The higher score in 2022 is an artifact of the algorithm grouping highly noisy, anomalous traffic into fewer, massive macro-clusters due to feature overlap. Conversely, the 2017 baseline contained 16 highly specific micro-clusters. The Silhouette metric mathematically penalized the 2017 score due to the close proximity of these micro-clusters in 78-dimensional space (the "Curse of Dimensionality"), despite them being visually distinct deterministic attacks.

5 Conclusions

The application of unsupervised machine learning to network telemetry reveals a clear contrast between academic datasets and real-world traffic. The 2017 baseline data, generated via controlled scripts, allowed DBSCAN to easily identify and isolate 16 distinct micro-clusters. However, when deployed against the 2022 LUFlow data, the algorithm struggled. Severe feature overlap and chaotic background radiation tricked the algorithm into forming massive macro-clusters, drastically reducing its ability to confidently flag anomalies (only 1.61% compared to 5.47% in the baseline).

This project mathematically and visually confirms that modern internet background noise and attacker evasion techniques severely degrade the efficacy of traditional spatial clustering.

Future iterations of this research should explore hybrid models, such as using Autoencoders for initial dimensionality reduction before clustering, to better handle the complexities of modern IDS deployments.

References

- [1] Sharafaldin, I., Lashkari, A. H., & Ghorbani, A. A. (2018). *Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization*. 4th International Conference on Information Systems Security and Privacy (ICISSP).
- [2] Mills, R., Broadbent, M., Marnerides, A., & Race, N. (2020). *LUFlow: A Flow-based Network Intrusion Detection Dataset*. Lancaster University.
- [3] Ester, M., Kriegel, H. P., Sander, J., & Xu, X. (1996). *A density-based algorithm for discovering clusters in large spatial databases with noise*. In Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD-96).